Quantitative Assessment of Galileo SAR Satellite System Resilience to Jamming Attacks

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***Abstract*—The Galileo Search and Rescue system is a European satellite-based service designed to locate individuals in distress. With a design life exceeding twelve years, its core components depend heavily on specific metrics, namely reliability and availability, to guarantee continuous operation and resilience. However, security threats, particularly jamming attacks, can severely compromise mission reliability. In this study, we model attacker and defender components based on the SPARTA attack behavior catalog. Metrics related to signal availability, attacks, and defense are incorporated into Continuous Time Markov Chains within the PRISM probabilistic model checker. Consequently, we conduct a rigorous quantitative analysis regarding the resilience within the integrated satellite system under an attack-defense scenario. Finally, these scenarios are simulated in GNU Radio to produce trustworthiness verdicts, which are then compared with the formal model verification results.**

***Index Terms*—Satellite, Reliability, Threats, PRISM**

I. Introduction

Satellite systems are present in various sectors, such as maritime [1] and military operations [2]. Given the critical nature of these systems, dependability and security are paramount, given the complex challenges involved [3] for search and rescue (SAR) missions, where human safety is the primary priority [4], [5], [6].

Formal verification [7], [8] is a mathematical technique that enables rigorous system analysis and the formal verification of properties, ensuring correct operation even in the presence of stochastic phenomena. Various formalisms, including Markov Decision Processes (MDPs), Continuous Time Markov Chains (CTMCs), and Concurrent Stochastic Games (CSGs), are tailored to specific applications. In contrast to simulation, which relies on analyzing results from many random samples, formal verification mathematically derives trustworthiness verdicts by evaluating system requirements expressed in formal logic syntax.

Our research illustrates how to effectively model communication efficiency and verify system performance using the PRISM model checker [8]. While PRISM is widely used to verify hardware and software designs [9], our methodology specifically addresses dependability and security by employing a Continuous Time Markov Chain (CTMC) representation [10]. Previous studies [11], [12], [13], [14] primarily focus on assessing the physical dependability of satellite hardware, such as the US GPS constellation. In contrast, this study investigates the communication link between the satellite and ground receivers, a segment highly susceptible to jamming [15]. Furthermore, we evaluate system resilience and the capacity to recover from interference-induced noise [16] by incorporating specific defense strategies to maintain operational efficiency. The frequency parameters governing attacks and defenses are treated as configurable variables, whereas the reliability metrics are derived from official satellite documentation [17]. These parameters are integrated into the formal model to assess overall system reliability in the presence of adversarial threats and countermeasures. Finally, the selected values are used in empirical GNU Radio [18], [19] experiments to generate trustworthiness verdicts, which are subsequently compared against the outcomes of the formal mathematical verification.

Outline: The remainder of this paper is organized as follows. Section II reviews the related work. Section III provides a brief background on CTMCs. Section IV details our proposed modeling approach for Galileo systems in the presence of attacks-defense scenario. In Section V, we perform a quantitative analysis and simulation using the GNU Radio platform. Finally, Section VI concludes the paper and outlines future research.

II. Related works

Several works in the literature have addressed the Performance and reliability of satellite systems. In [13], [11], the authors propose using CTMCs to model satellite systems, demonstrating how this approach can analyze the impact of factors such as solar radiation on satellite reliability and maintenance, as well as on scrubbing activities. Additionally, [14] broadens the scope by incorporating human capabilities in maintaining satellite systems within a game-theoretical model.

Several surveys have focused on space cybersecurity, notably [3], [20]. In parallel, other studies, such as [21], have focused on developing security metrics for space systems by building upon the NIST framework. Authors

in [3] investigate potential space attack vectors encompassing the ground segment, space segment, user links, cloud communication channels, and supply chains. They propose fostering new standards and guidelines to fortify space missions, outlining mitigation strategies that rely on classical methods such as intrusion detection systems, while also exploring quantum computing for threat detection. Similarly, authors in [20] provide a comprehensive survey of the cybersecurity landscape surrounding satellite communications (Satcoms). They review both general cybersecurity strategies and specific technical mechanisms designed to defend Satcom systems against cyberattacks. Finally, the authors in [21] propose new metrics to assess the performance of satellite systems under attack. They argue that although such performance metrics are not currently included in the NIST framework, they can be augmented with specific indicators, such as the risk and frequency of signal-jamming attempts in space communications.

While most existing approaches focus primarily on the technological specifications and performance metrics of satellite components, they often overlook vulnerabilities exposed by jamming attacks within stochastic models [3], [20], [21]. Moreover, because reliability and security threats are usually treated separately, the existing literature lacks a compositional approach that integrates both attributes. Our methodology employs a hybrid approach. First, we model the attack-defense scenario in PRISM, incorporating parameters for satellite systems and specific attack and defense rates. Subsequently, we integrate these metrics into a GNU Radio simulation platform. This combined approach allows us to evaluate the practical gains of the defense mechanisms during simulation and directly compare them with the formal verification results.

III. Preliminaries

Probabilistic Model Checking with PRISM is a well-established modeling and verification technique for evaluating reliability and performance. It supports various formalisms; for the specific requirements of our analysis, we employ Continuous-Time Markov Chains (CTMCs) [22] to model the reliability of satellite systems under attack scenarios. The CTMC involves a set of states and a transition matrix $\mathbf{R} : S \times S \to \mathbb{R}_{\geq 0}$. The rate specifies the delay before a transition between states s and s' takes place $\mathbf{R}(s, s')$, where the probability between s and s' take within time t is given by the value $1 - e^{-\mathbf{R}(s,s') \times t}$. Based on research using PRISM for CTMC modeling, as outlined in [23], exponentially distributed delays are often considered suitable for modeling electronic component lifetimes and inter-arrival times. Since the satellite has no prior knowledge of the attacker's schedule, the jamming is considered a random event uncorrelated with the system's internal state.

The PRISM model comprises modules that can synchronize. Each module is characterized by variables and commands (or transitions). The values of these variables reflect the module's state. The behavior of each module is described using a set of commands, each of which follows the following format: $[a]\ g\ \rightarrow\ \lambda : u$

This indicates that if the guard condition g evaluates to true, then the update u is enabled to occur with a rate of λ for action a.

Properties are expressed in Continuous Stochastic Logic (CSL) [22]. For instance, the following property described in natural language: *What is the probability that the distress signal will be lost for a period longer than 100 minutes during a jamming attack?* is expressed as: $P_{<0.001}[F^{\leq 100}\ fail]$ Here, $fail$ is the label that refers to the system failure states.

IV. The SAR/Galileo system

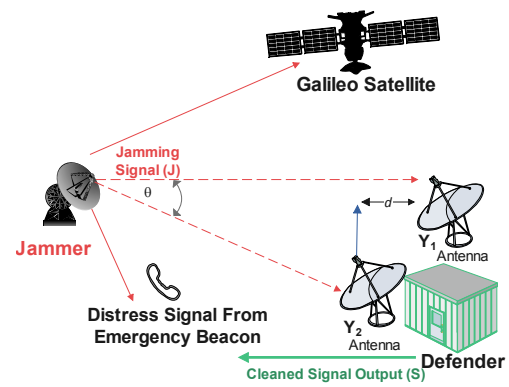

Figure 1: SAR/Galileo Services [4], [5].

COSPAS SARSAT (C/S) is an international satellite-based Search and Rescue (SAR) distress alerting system established in 1979 by the USA, Canada, France, and the former USSR [17] (see Figure 1). The full documentation is available in [24]. The system encompasses multiple components that relay beacon signals from distressed users to rescue authorities. We model a scenario in which adversaries conduct jamming attacks on communication links. In these cases, the attacker employs jammers emitting signal noise to tamper with or compromise the integrity of the desired satellite signal. According to [4] and [5], Galileo satellites demonstrate a reliability exceeding 88% over a 12-year lifespan, with an availability of 99.5%. The work in [6] reports that the availability of healthy Galileo signals is 99.22%, with a recovery time of less than 15 hours.

Jamming constitutes the intentional emission of radio frequency noise to interfere with communications between space segment and ground stations according to SPARTA catalog [25]. However, satellites can be designed with antennas (on ground segment, see Figure 1) that "null" signals from a particular geographic region on the surface of the Earth or from locations in space where jamming is detected [26]. In our scenario, a jammer emits noise J(t to interfere with satellite signals intended for a distressed user (*e.g.*, on a vessel or aircraft), potentially degrading the link quality. This jamming signal consists of high-power Gaussian white noise modulated by a 1 Hz pulse.

A. Formal modeling

The global variables within the PRISM model, presented in Listing IV.1 (lines 3-19), include variables related to different states of the satellite and signal health in lines 3 to 4, alongside attacking and defense states in lines 6-9. Reliability parameters that capture multiple failure origins (λ) and recovery rates (μ), such as those related to the signal failure rate (λ_{signal}) and signal recovery (μ_{signal}), are located in lines 12 to 15. The jamming and nulling variables are modeled in lines 14 to 19. For instance, λ_{jam} refers to the *mean time to perform an attack*, and μ_{jam} is the *mean time between attacks* (i.e., sleeping time). Moreover, λ_{null} indicates the *mean time to perform a defense*, whereas μ_{null} represents the *mean time between defenses* (sleeping time).

The PRISM modules, outlined in lines 22-44, formally represent the degradation of the satellite and signal health. They model the transition from an operational state to a degraded mode and the subsequent recovery to full operation. The "Attack" module captures adversarial behavior through two primary state transitions. The first transition shifts the system from a standby state to a jammed state at a rate of λ_{jam}, while the second represents the recovery from the jammed state back to an idle state. Similarly, the defense module models corresponding state transitions, moving from an idle state to an active defense state (*i.e.*, nulling), at a rate of λ_{null}, and returning from the nulling operation to the idle state at a rate of μ_{null}. Finally, the formula defined in line 45 expresses the overall operational status of the system. This is formulated as a logical conjunction of the satellite's operational status and the signal, requiring either the absence of an attack or the simultaneous execution of a nulling operation during an active attack.

Listing IV.1: The Service Status Over Execution Time

```
1  ctmc
2  // System state constants
3  const int OPERATIONAL = 2; // Component is functional
4  const int DEGRADED    = 1;// Component has failed
5  // states for Attack/Defense context
6  const int NO_ATTACK   = 2;// No jamming detected
7  const int JAMMED      = 1;// System is under attack
8  const int DEF_IDLE    = 2;// Defense is ready
9  const int NULLING     = 1;// Defense is active
10
11 // Performance rates
12 const double λ; // Failure rate based on MTBF
13 const double μ; // Repair rate based on MTTR
14 const double λ_signal; // Signal Failure rate
15 const double μ_signal; // Signal recovery rate
16 const double λ_jam; //Rate of jamming occurrence
17 const double mu_jam; // Rate of jamming end
18 const double λ_null ; // Rate of activating nulling
19 const double μ_null ; // Rate of defense reset
20
21 //Satellite Degradation Module
22 module Satellite_Degradation
23  SAT : [0..2] init OPERATIONAL;
24  [sat_degradation] SAT>0        → λ:(S'=Degraded);
25  [sat_recovery]    SAT=Degraded→ μ:(S'=OPERATIONAL);
26 endmodule
27 // Signal Degradation Module
28 module Signal Degradation Module
29  SIG : [0..2] init OPERATIONAL;
30  [sig_degradation] SIG > 0 → λ_signal:(SIG'=Degraded);
31  [sig_recovery] SIG = DEGRADED → μ_signal:(SIG'=
       OPERATIONAL);
32 endmodule
33 //Attack Module
34 module Attack
35  ATT : [0..2] init NO_ATTACK;
36  [jamming] ATT > 0 -> → λ_jam : (ATT' = JAMMED);
37  [jamming_rec] ATT = JAMMED -> → μ_jam : (ATT' =
       NO_ATTACK);
38 endmodule
39 //Defense Module
40 module Defence
41  DEF : [0..2] init DEF_IDLE;
42  [nulling] DEF > 0 -> → λ_null : (DEF' = NULLING);
43  [nulling_rec] DEF = NULLING -> → μ_null : (DEF' =
       DEF_IDLE);
44 endmodule
45 formula system_up = (SAT = OPERATIONAL) & (SIG =
      OPERATIONAL) & ( (ATT = NO_ATTACK) | (ATT =
      JAMMED & DEF = NULLING) );
```

B. GNU Radio-based simulation

To model signal reception, we assume the defender uses a two-antenna array producing outputs Y_1 and Y_2. These elements are separated by a distance d relative to an angle of incidence θ. For a Uniform Linear Array (ULA) [27], the phase shift can be defined as $\Delta\phi(\theta) = \pi \sin(\theta)$. Consequently, the first antenna receives the superposition of the satellite signal and the jamming interference at time instant t:

$$Y_1(t) = S_{sat}(t) + J_{jam}(t) \tag{1}$$

Similarly, the signal received at the second antenna incorporates the phase shift relative to the incident angle:

$$Y_2(t) = S_{sat}(t) + J_{jam}(t) \times e^{j\Delta\phi(\theta_{jam})} \tag{2}$$

We assume that the satellite signal arrives at a broadside angle ($\theta_{sat} = 0°$); therefore, it incurs no phase shift (where j represents the imaginary unit from Euler's formula).

A defender located in the same geographical region can employ nulling techniques to suppress the jamming signal while recovering a usable signal for the user. The defender applies a complex correction weight [28] $W = e^{-j\Delta\phi(\theta_{est})}$ to compute the output $Z(t) = Y_1(t) - Y_2(t) \times W$. This operation yields the following:

$$\begin{aligned} Z(t) = {} & S_{sat}(t) \times (1 - e^{-j\Delta\phi(\theta_{estim})}) \\ & + J_{jam}(t) - J_{jam}(t) \times (e^{j\Delta\phi(\theta_{jam})} \times e^{-j\Delta\phi(\theta_{est})}) \end{aligned} \tag{3}$$

If the estimated angle matches the jamming angle, meaning $\theta_{est} = \theta_{jam}$, then the equation simplifies to:

$$Z(t) = S_{sat}(t) \times (1 - e^{-j\Delta\phi(\theta_{est})}) \tag{4}$$

This result confirms the elimination of $J(t)$ (Nulling). Under these conditions, the defender successfully isolates the satellite signal.

For experimental validation, we recorded simulation execution using the GNU Radio platform to monitor successful attacks and defenses. Thus, we can reason about the model's resilience regarding attacks.

V. Quantitative analysis using PRISM

In this work, we evaluate the efficacy of attack and defense strategies from the SPART catalog using the PRISM tool. To validate these models, we deploy a radio-based simulation using GNU Radio.

a) Experimental setup.: We encoded properties in CSL using the PRISM model checker (v4.8) for verification. These experiments were conducted on an Ubuntu system with an i7 processor and 32GB of RAM. The healthy signal failure is $\lambda_{signal} = 0.001$ and the repair rate $\mu_{signal} = 8.7 \times 10^{-6}$. Using the configuration, including parameters related to the healthy signal, we set the jamming rate $\lambda_{jam} = 1/MTTA$ with off-period $\mu_{jam} = 1/2$; however, the nulling rate $\lambda_{null} = \alpha/MTTRA$ with off-period $\mu_{null} = 1/2$. According the documentation, the satellite failure rate is $\lambda = 2.03 \times 10^{-8}$ and repair rate is $\mu = 4.03 \times 10^{-6}$. Then, we built an initial setup using GNU Radio v3.10.1.1 that comprises a satellite generating the source signal, an attacker performing jamming, and a defender responsible for signal recovery. The satellite is aligned at 0 degrees relative to the defender's first antenna, whereas the attacker is positioned at 45 degrees relative to the second antenna.

b) Artifacts.: The source code for the experiments, including the GNU Radio architecture and PRISM code, is available on a GitHub repository [29].

A. Model checking

$$\mathtt{P} = ?[(\text{“up”})\ \mathtt{U}^{\leq \mathtt{T}}\ !(\text{“up”})] \qquad (\text{“Degraded”})$$

During the PRISM verification process, the mean time to mitigate a threat is calculated relative to the mean time to perform an attack (MTTA). By treating the defense rate as a constant multiplier of the attack parameters, we can observe a direct correlation between attack frequency and the corresponding defenses (see Figure 2). Property *“Degraded”* demonstrates that the impact of the attack diminishes as time elapses. As the jamming rate λ_{jam} decreases, its effect on the system becomes less significant. In scenarios where the defense factor equals $\alpha = 1$, the response curves appear parallel (-0.01). However, when the defense factor increases so that the defense rate exceeds the attack rate ($\lambda_{nulling} = \alpha \times \lambda_{jam}$), the system yields a significant mitigation, making the configuration more resilient to attacks. Upon evaluating the reward function associated with this defense mechanism, we determined that the expected reward is 4.26% ($\alpha = 10$). This specific outcome is a direct consequence of the selected defense rate and the attack frequency.

B. Simulation-based validation

Following the formal verification performed via the PRISM model checker, we validate our theoretical approach through empirical simulation. For this purpose, we assign constant values to the system parameters: the jamming attack rate is set to $\lambda_{jam} = 1/100$ with a recovery rate of $\mu_{jam} = 1/2$, while the nulling defense rate is defined as $\lambda_{null} = 1/10$ with a corresponding recovery rate of $\mu_{null} = 1/2$. Using these fixed parameters, we execute the simulation to assess the overall system rewards derived from operating defenses against active attacks. We configure the model to adhere to the rates defined previously. Calculations are performed on the data collected and recorded within the generated GNU Radio binary file. Specifically, we compute the ratio of successfully defended signals to the total number of signals originating from the satellite. This empirical evaluation yields a reward value of 5.85%, which aligns with our prior formal verification results. The observed disparity of 1.59% is not a contradiction but rather quantifies the gap between probabilistic theory and physical-layer simulation. While PRISM establishes a stochastic lower bound (worst-case scenario), the simulation demonstrates the system's nominal performance under a periodic (1Hz) attack period.

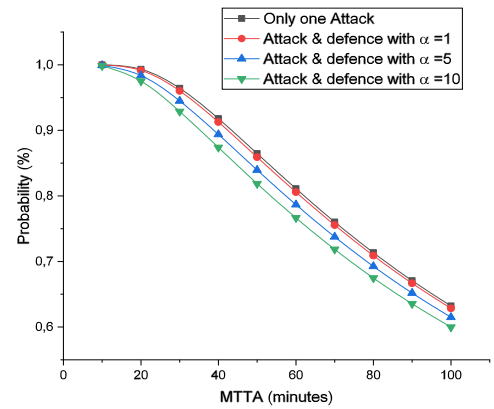

Figure 2: Attacks defences mitigation effects.

C. Discussion & Threats to validity

The results demonstrate that reliability and availability are not the only parameters influencing satellite performance over the 12-year operational period. Attacks that explore vulnerabilities in satellite communication can compromise signal accuracy. While the operational activity described in [13], [11] relies exclusively on standard reliability metrics, our findings confirm that malicious attacks constitute a legitimate and significant threat to mission success. We have constrained our experimental setup to a single attacker and a defender equipped with a dual antenna array. However, the SPARTA catalog contains numerous additional attack vectors that can be implemented. Furthermore, satellite performance metrics can be expanded to include additional parameters, such as orbital trajectory and real-time position updates. Finally, the attacker's schedule is modeled as a memoryless exponential process, whereas in real cases it can follow a phase-type distribution, which yields a more accurate approximation for non-exponential attack patterns.

VI. Conclusion

This paper presents a stochastic approach using CTMCs to model SAR/Galileo communication services under jamming attacks that impede signal transmission. We explicitly model a jamming attack alongside potential defense mechanisms on satellite-to-ground links, using PRISM to quantify the gains from defense scenarios and satellite dependability parameters. These parameters are also implemented in the GNU Radio simulation platform to compute the gains of system resilience resulting from the defenses' effects, and the results are compared with the formal verification results. For future activities, to mitigate the variance of the exponential distribution, we will implement an Erlang distribution for the jamming duration by splitting the jamming state into two sequential intermediate phases.

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